MSG No. 121.332 $I_c\bar{4}2m$ [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `-42m`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'2m'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,3,4,29,30,31,32]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[5,6,7,8,25,26,27,28]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,10,11,12,21,22,23,24]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20]`

Table 3: Wyckoff site: `4c`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,13,14,18,19,31,32]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,15,16,17,20,29,30]`
3	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,6,9,12,23,24,26,27]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[7,8,10,11,21,22,25,28]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,4,7,8,18,19,21,22]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,3,5,6,17,20,23,24]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,12,15,16,26,27,29,30]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[10,11,13,14,25,28,31,32]`

Table 5: Wyckoff site: `8e`, site symmetry: `2.mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,7,8]`
2	$[0,\ 0,\ -z]$	`[2,3,5,6]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12,15,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,11,13,14]`
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[17,20,23,24]`
6	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[18,19,21,22]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[25,28,31,32]`

8	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[26,27,29,30]

Table 6: Wyckoff site: 8f, site symmetry: 2.m'm'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,4,31,32]
2	$\left[0,\ \frac{1}{2},\ -z\right]$	[2,3,29,30]
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	[5,6,26,27]
4	$\left[\frac{1}{2},\ 0,\ z\right]$	[7,8,25,28]
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[9,12,23,24]
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[10,11,21,22]
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[13,14,18,19]
8	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,16,17,20]

Table 7: Wyckoff site: 8g, site symmetry: 2.mm

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[1,7,28,32]
2	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	[2,6,27,29]
3	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	[3,5,26,30]
4	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[4,8,25,31]
5	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[9,15,20,24]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[10,14,19,21]
7	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[11,13,18,22]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[12,16,17,23]

Table 8: Wyckoff site: 16h, site symmetry: .2.

No.	position	mapping
1	$[0,\ y,\ 0]$	[1,3]
2	$[0,\ -y,\ 0]$	[2,4]
3	$[y,\ 0,\ 0]$	[5,8]
4	$[-y,\ 0,\ 0]$	[6,7]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[9,11]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[10,12]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,16]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}\right]$	[14,15]
9	$\left[0,\ y,\ \frac{1}{2}\right]$	[17,19]
10	$\left[0,\ -y,\ \frac{1}{2}\right]$	[18,20]
11	$\left[y,\ 0,\ \frac{1}{2}\right]$	[21,24]
12	$\left[-y,\ 0,\ \frac{1}{2}\right]$	[22,23]
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	[25,27]
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[26,28]
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[29,32]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ 0\right]$	[30,31]

Table 9: Wyckoff site: `16i`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,19]`
2	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[2,20]`
3	$\left[0,\ y,\ \frac{3}{4}\right]$	`[3,17]`
4	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[4,18]`
5	$\left[y,\ 0,\ \frac{3}{4}\right]$	`[5,24]`
6	$\left[-y,\ 0,\ \frac{3}{4}\right]$	`[6,23]`
7	$\left[-y,\ 0,\ \frac{1}{4}\right]$	`[7,22]`
8	$\left[y,\ 0,\ \frac{1}{4}\right]$	`[8,21]`
9	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[9,27]`
10	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[10,28]`
11	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[11,25]`
12	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[12,26]`
13	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[13,32]`
14	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[14,31]`
15	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[15,30]`
16	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[16,29]`

Table 10: Wyckoff site: `16j`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,8]`
2	$[x,\ -x,\ -z]$	`[2,5]`
3	$[-x,\ x,\ -z]$	`[3,6]`
4	$[-x,\ -x,\ z]$	`[4,7]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,16]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[10,13]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,14]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[12,15]`
9	$\left[x,\ x,\ z+\frac{1}{2}\right]$	`[17,24]`
10	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	`[18,21]`
11	$\left[-x,\ x,\ \frac{1}{2}-z\right]$	`[19,22]`
12	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	`[20,23]`
13	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[25,32]`
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	`[26,29]`
15	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -z\right]$	`[27,30]`
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	`[28,31]`

Table 11: Wyckoff site: `16k`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	`[1,32]`
2	$\left[x,\ \frac{1}{2}-x,\ -z\right]$	`[2,29]`
3	$\left[-x,\ x+\frac{1}{2},\ -z\right]$	`[3,30]`
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	`[4,31]`

continued ...

Table 11

No.	position	mapping
5	$\left[x+\frac{1}{2},\,-x,\,-z\right]$	[5,26]
6	$\left[\frac{1}{2}-x,\,x,\,-z\right]$	[6,27]
7	$\left[\frac{1}{2}-x,\,-x,\,z\right]$	[7,28]
8	$\left[x+\frac{1}{2},\,x,\,z\right]$	[8,25]
9	$\left[x+\frac{1}{2},\,x,\,z+\frac{1}{2}\right]$	[9,24]
10	$\left[x+\frac{1}{2},\,-x,\,\frac{1}{2}-z\right]$	[10,21]
11	$\left[\frac{1}{2}-x,\,x,\,\frac{1}{2}-z\right]$	[11,22]
12	$\left[\frac{1}{2}-x,\,-x,\,z+\frac{1}{2}\right]$	[12,23]
13	$\left[x,\,\frac{1}{2}-x,\,\frac{1}{2}-z\right]$	[13,18]
14	$\left[-x,\,x+\frac{1}{2},\,\frac{1}{2}-z\right]$	[14,19]
15	$\left[-x,\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$	[15,20]
16	$\left[x,\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$	[16,17]

Table 12: Wyckoff site: 321, site symmetry: 1

No.	position	mapping
1	$[x,\,y,\,z]$	[1]
2	$[x,\,-y,\,-z]$	[2]
3	$[-x,\,y,\,-z]$	[3]
4	$[-x,\,-y,\,z]$	[4]
5	$[y,\,-x,\,-z]$	[5]
6	$[-y,\,x,\,-z]$	[6]
7	$[-y,\,-x,\,z]$	[7]
8	$[y,\,x,\,z]$	[8]
9	$\left[x+\frac{1}{2},\,y+\frac{1}{2},\,z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,\frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,\frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\,\frac{1}{2}-y,\,z+\frac{1}{2}\right]$	[12]
13	$\left[y+\frac{1}{2},\,\frac{1}{2}-x,\,\frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-y,\,x+\frac{1}{2},\,\frac{1}{2}-z\right]$	[14]
15	$\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$	[16]
17	$\left[x,\,y,\,z+\frac{1}{2}\right]$	[17]
18	$\left[x,\,-y,\,\frac{1}{2}-z\right]$	[18]
19	$\left[-x,\,y,\,\frac{1}{2}-z\right]$	[19]
20	$\left[-x,\,-y,\,z+\frac{1}{2}\right]$	[20]
21	$\left[y,\,-x,\,\frac{1}{2}-z\right]$	[21]
22	$\left[-y,\,x,\,\frac{1}{2}-z\right]$	[22]
23	$\left[-y,\,-x,\,z+\frac{1}{2}\right]$	[23]
24	$\left[y,\,x,\,z+\frac{1}{2}\right]$	[24]
25	$\left[x+\frac{1}{2},\,y+\frac{1}{2},\,z\right]$	[25]
26	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,-z\right]$	[26]
27	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\,\frac{1}{2}-y,\,z\right]$	[28]
29	$\left[y+\frac{1}{2},\,\frac{1}{2}-x,\,-z\right]$	[29]

continued ...

Table 12

No.	position	mapping
30	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, -z\right]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[32]